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Linear programming produces greater, earlier and uninterrupted neuromuscular and functional adaptations than daily-undulating programming after velocity-based resistance training

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ABSTRACT

This study aimed to compare the effect of linear (LP) and daily-undulating (DUP) programming models on neuromuscular and functional performance using the velocity-based resistance training (VBRT) approach. Thirty-two resistance trained men were randomly assigned into 2 groups: LP ($n = 16$) or DUP ($n = 16$). Both training groups completed an 8-week VBRT intervention using the full squat exercise, only differing in the relative intensity (% 1RM) distribution during the training program. Changes produced by each periodization model were evaluated using the following variables: estimated 1RM; average mean propulsive velocity (MPV) attained for all absolute loads common to Pre-test and Post-test; average MPV attained against absolute loads lifted faster than 1 $m{\bullet}s^{-1}$; average MPV attained against absolute loads lifted slower than 1 $m{\bullet}s^{-1}$; countermovement jump (CMJ) and fatigue test. Moreover, CMJ and 1RM parameters were evaluated weekly to analyze their evolution along the training program. LP and DUP strategies significantly improved all performance variables analyzed ($p<0.001$), except the fatigue test in the DUP group. Significant "time x group" interactions were observed in all strength variables and fatigue test in favour of the LP group. In addition, pre-post effect size (ES), percentages of change and weekly comparisons showed higher improvements in the LP group (ES=0.54–2.49, Δ=9.5–60.4%) compared to DUP (ES=0.40–1.65, Δ=5.5–27.2%). Based on these findings, the LP appears to stand as a more effective strategy than DUP to achieve greater, earlier and uninterrupted neuromuscular and functional adaptations in VBRT interventions.

1. Introduction

Neuromuscular adaptations induced by resistance training (RT) directly depend on the structure of the training program [1,2]. Configuration of training stimuli involve the manipulation of different specific variables to obtain the targeted goals [1]. Traditionally, these key variables include: 1) relative load (%1RM), 2) volume, 3) training frequency, 4) exercise type and order, 5) rest intervals between repetitions, sets and exercises, 6) movement velocity, and 7) type of muscle activation [1,3]. Whereas the influence of the aforementioned variables has been widely analyzed in the scientific literature [1,2,4], the effect of other variables, including the configuration (evolution) of relative load and volume during the RT program [5-7] has been less addressed. This factor has been commonly referred to as periodization, although the appropriate term to define the variation of relative load and volume during a training cycle is programming [8].

In several studies have been found a superior effectiveness of scheduled RT compared to non- scheduled RT to improve physical performance, regardless of training volume or training status of participants [7,9]. However, despite the widely supported superiority of scheduled RT interventions, the type of configuration (i.e., programming) that induces the greatest improvements on strength gains remains still unknown. Although a myriad of relative load and volume configurations exist, the two commonly used forms of RT programming models

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used in longitudinal studies have been the linear (LP) and the daily-undulating (DUP) approaches [5,10,11]. Briefly, the LP model is based on a gradual increase of relative load while training volume is reduced (not necessarily in every training session) as RT progresses, whereas in the DUP model, training volume and relative load are varied on a daily basis [10,12].

Traditionally, RT programs have been configured using the LP strategy [11,13]. However, two main disadvantages have been related to this model: i) the fatigue state produced by the continuous increase of relative load during the cycle, and ii) the performance stagnation generated by lengthy (i.e., duration) training periods spent in a specific relative load [11]. For these reasons, some authors have indicated that the DUP model may produce greater strength improvements since, by incorporating more frequent variations in relative load, the athletes may be forced to continually adapt to the unfamiliar stress (i.e., new training stimulus in each training session) [10-12]. Thus, there would be stress optimized and recovery on the neuromuscular components [13]. Indeed, it has been proclaimed that variations in training stimulus appear to be of paramount importance for increasing strength gains [7]. According to this assertion, some studies comparing different programming models have shown greater muscle strength gains for DUP compared to LP, in both the upper- and lower-limb exercises [12,14-17]. However, most studies in this matter have found similar improvements in strength performance, hypertrophy and local muscular endurance for both types of programming strategies [5,18-26], and even greater improvements for LP than DUP models [27,28]. Therefore, it appears that no clear evidence in favor of either of the two programming approaches (LP or DUP) have been found for the development of upper or lower body strength.

An important aspect to compare the neuromuscular adaptations induced by RT programs based on DUP and LP models is that the rest of the variables (i.e. relative loads, volume, exercises, inter-set rest and frequency) remain equaled between both training protocols. However, in several previous studies [20,21,27,29] the LP and DUP programs were conducted using different relative load and volume. Therefore, it is not known if the results of these studies were due to the different training configurations analyzed (i.e., programming strategy) or other confounding factors. In addition, studies matching the relative load and volume between LP and DUP during the training period [15-19,22,28] present two important concerns in relation to the control of the training stimulus and the improvement of sports performance. Firstly, training load has been commonly established according to an initial evaluation with no readjustments during the training period. Considering that the strength gains are probably different for LP and DUP models throughout the training period, we cannot be sure that both programming models were performed with the same relative load. In this regard, it has been indicated that monitoring movement velocity during RT represents an objective, valid and accurate method to determine both the relative load and volume during RT [30-33]. Secondly, conventional research comparing LP and DUP approaches have based their training configurations on sets to muscle failure [12,16,18,20,22,28,34]. However, several previous studies [35-38] have shown that performing a moderate number of repetitions per set (half or less than half the maximum number of repetitions that could be performed per set leading to failure) induces greater enhancements in strength, muscle power, and other motor skills (e.g., rowing, jumping, sprinting) compared to repetitions to failure in trained and untrained individuals.

Finally, most of the studies comparing LP vs. DUP have focused on changes in muscle strength and hypertrophy [5,10,11,13], whereas only a few studies have analyzed the effectiveness of both programming models on other important muscle abilities, such as the vertical jump [14,21,39] and local muscle endurance [18,28]. Therefore, based on the contradictory results found in previous comparisons of LP and DUP strategies, added to the aforementioned methodological limitations of traditional investigations, the aim of the present study was to compare the effect of LP and DUP models, using velocity-based squat training on strength, local muscle endurance and jump performance in physically active young men.

2. Materials and methods

2.1. Study design

A longitudinal and experimental study was used to analyze the effects of manipulating the evolution of the relative load (% 1RM) during a training period (LP vs. DUP) on strength and jump performance. To address this, 32 healthy men were randomly allocated into two groups: LP ($n = 16$) and DUP ($n = 16$). All participants were assessed before (Pre) and after (Post) the training period using a countermovement jump (CMJ) test and a progressive loading test in the full-squat (SQ) exercise. In addition, CMJ and one-repetition maximum (1RM) parameters were evaluated weekly to analyze their evolution along the training program. Following the initial evaluation, both experimental groups trained twice a week (72 h rest) during an 8-week training period using only the SQ exercise. In the present study, the characteristics of the RT program for both LP and DUP were exactly the same. The only difference was the way the relative load during the training cycle was programmed: one group carried out progressively greater load magnitudes training (i.e., LP training), whereas the other group conducted RT in which the load magnitude varied daily (i.e., DUP training). All training sessions were conducted in a research laboratory under the direct supervision of the investigators and under controlled environmental conditions (~21 °C and ~60% humidity).

2.2. Participants

Thirty-two healthy men volunteered to take part in this study. Participants were physically active sport science students. Inclusion criteria were: i) being capable of performing a technically correct squat exercise; and ii) being injury free for at least 8 months before participating in this study; iii) do not participate in any other type of strenuous physical activity during the time this investigation lasted; iv) having a RT experience ranging from 1 to 3 years (with at least 1–3 sessions per week); v) do not take drugs, medications, or dietary supplements known to influence physical performance or hormonal balance before or during this study; vi) do not have physical limitations, disease or health problems that could affect the testing or training sessions. After initial evaluation, participants were matched according to their estimated 1RM in the SQ exercise and then randomly assigned into two groups: a linear programming group (LP, $n = 16$) and a daily-undulatory programming group (DUP, $n = 16$). Flowchart of study participants is shown in Fig. 1 and characteristics of participants who completed the intervention are displayed in Table 1. The study was conducted according to the Declaration of Helsinki and was approved by the Research Ethics Committee of Pablo de Olavide University (Authorization code: EH-1/2015). After being informed of the purpose and experimental procedures, the participants signed a written informed consent form prior to participation.

2.3. Testing procedures

All evaluations were performed in a single testing session. Participants arrived to the laboratory in a well-rested condition and underwent a medical examination and anthropometric assessments. Then, each participant performed a CMJ test, a progressive loading test and a fatigue test in the SQ exercise. Before the physical performance assessment, all participants carried out a general standardized warm-up which consisted of 5 min of running at a self-selected intensity, 5 min of joint mobilization exercises, followed by 3 sets of progressively faster 30 m running accelerations. Exactly the same testing protocol was carried out during both Pre and Post testing sessions. At least two experienced researchers supervised the testing session to ensure correct and consistent techniques during all tests. Strong verbal encouragement was provided

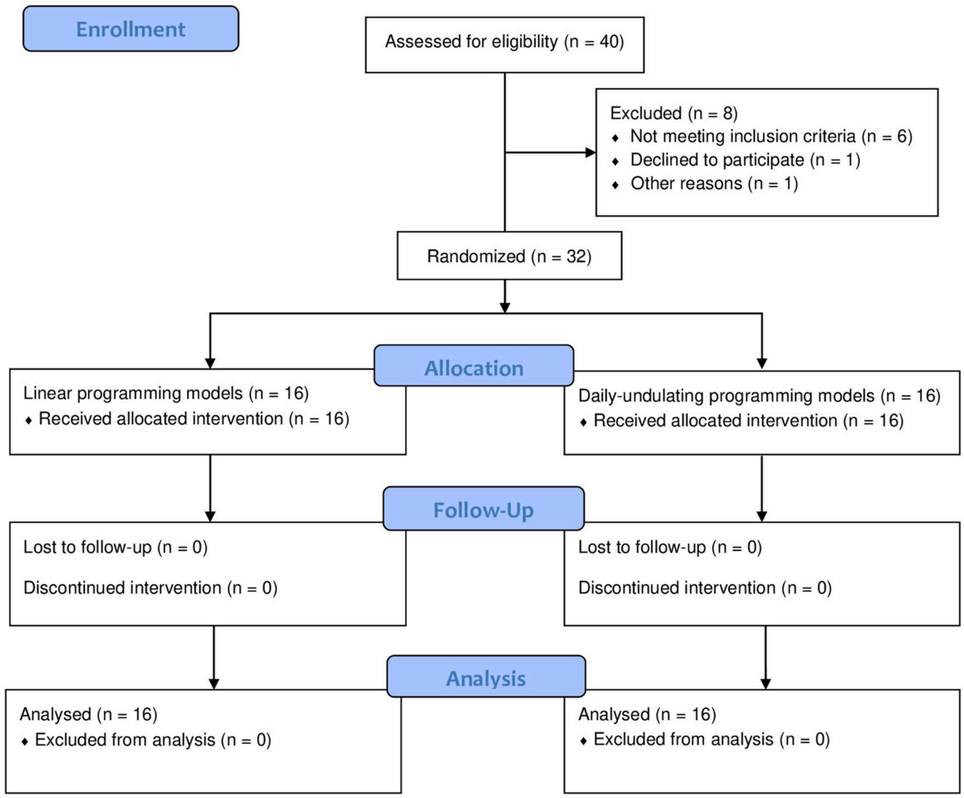

Fig. 1. Flowchart of participants included in the study.

Table. 1
Participant's physical characteristics.

Group	Age (years)	BM (kg)	Height (m)	BF (%)	MM (%)	RSR
LP	24.3 ± 4.2	75.5 ± 9.1	1.76 ± 0.06	12.2 ± 5.9	57.3 ± 11.8	1.33 ± 0.26
DUP	22.4 ± 2.7	76.1 ± 8.1	1.78 ± 0.07	13.6 ± 5.8	57.1 ± 10.3	1.33 ± 0.25

Data are mean ± SD. LP: Linear programming group; DUP: Daily-undulatory programming group; BM: Body mass; BF: Body fat; MM: Muscle mass; RSR: Relative strength ratio (1RM strength/body mass ratio).

during all assessments to motivate participants to give a maximal effort.

2.3.1. Countermovement jump test

A CMJ was performed with the participant standing in an upright position on an infrared timing system (Optojump System, Microgate, Bolzano, Italy) with the hands on the hips to avoid arm swings. From this position, participants performed a fast downward movement followed by a fast upward vertical movement as high as possible, all in one sequence, trying to reach the maximum possible height. Participants were instructed to land in an upright position and to bend the knees after landing. Flight time (t) and acceleration due to gravity (g) were used to calculate the vertical jump height (h) of the center of gravity of the body as follows: $h = t^2 \times g/8$. To ensure that the jump was completely vertical, the subjects were required to take off and land within the same area. If the subject did not take off or land within the designated area, the jump was not considered [40]. Five trials were completed with a 45 s rest between each trial. The highest and lowest jump height values were discarded, and the resulting mean value was kept for analysis [41]. Before trials, a specific warm-up consisted of 2 sets of 10 repetitions in the SQ exercise without extra load (2 min rest), 5 CMJ at progressive intensity (20 s rest) and 3 maximal CMJ (30 s rest). After 2-min rest, the test was carried out. The coefficient of variation (CV) for test-retest reliability was 1.15%, and the intraclass correlation coefficient (ICC) was 0.97 (95% CI: 0.94–0.99). The CMJ was measured at Pre, Post and before the second training session of each week, always using the same protocol described above, in order to analyze the evolution of this variable throughout the training program.

2.3.2. Progressive loading test in the SQ exercise

A detailed description of the SQ testing protocol has recently been provided elsewhere [32]. Testing was performed on a Smith machine (Multipower Fitness Line, Peroga, Murcia, Spain). The participants performed the SQ from an upright position, descending (eccentric phase) in a continuous motion until the posterior thighs and calves made contact with each other. Then, immediately reversed motion and ascended back to the starting position. The velocity of the eccentric phase was carefully controlled so that it always ranged between ~0.50–0.60 $m{\bullet}s^{-1}$, whereas participants were required to always execute the concentric phase of each repetition at maximal intended velocity. The individual position for the SQ exercise (feet position and placement of the hands on the bar) was measured for each participant so that they could be reproduced in all testing and training sessions. Specific warm-up consisted of two sets of eight and six repetitions (3-min rests) with loads of 20 and 30 kg, respectively. Initial load was set at 20 kg for all participants and was gradually increased in 10 kg increments until the mean propulsive velocity (MPV) was lower than ~0.60 $m{\bullet}s^{-1}$, which corresponds to ~85% 1RM [32]. Subjects performed a total of 7.8 ± 1.7 and 8.5 ± 1.6 increasing loads in the progressive loading test during Pre and Post, respectively. During the test, three repetitions were executed for light (MPV > 1.10 $m{\bullet}s^{-1}$), two for medium (1.10 $m{\bullet}s^{-1}$ > MPV > 0.80 $m{\bullet}s^{-1}$), and only one for the heaviest (MPV < 0.80 $m{\bullet}s^{-1}$) loads. Interset rests ranged from 3 (light) to 5 min (heavy loads). The exact same warm-ups and progression of absolute loads (kg) were repeated in the Post for each participant. Only the best repetition at each load, according to the criteria of fastest MPV, was considered for subsequent analysis. A linear velocity transducer (T-Force System;

Ergotech, Murcia, Spain) was used to register bar velocity. The following variables derived from this test were used for analysis: a) The 1RM calculated for each individual from the MPV attained against the heaviest load (kg) lifted in the progressive loading test, as follows: $(100 \times \text{load})/(-5.961 \times \text{MPV}^2) - (50.71 \times \text{MPV}) + 117$ [32]; b) average MPV attained against all absolute loads common to Pre and Post (AV) [42]; c) average MPV attained against absolute loads common to both tests that were lifted faster than 1 $m{\bullet}s^{-1}$ (AV > 1); and d) average MPV attained against absolute loads common to both tests that were lifted slower than 1 $m{\bullet}s^{-1}$ (AV < 1).

2.3.3. Fatigue test

Five minutes after finishing the progressive loading test, a fatigue test was performed against an absolute load that the participants could initially move to ~1.00 $m{\bullet}s^{-1}$ (~60% 1RM). Thus, before starting the test, adjustments in the load (kg) to be used were made when needed, so that the velocity of the first repetition matched the specified target MPV. During each test, the participants were required to move the bar as fast as possible during the concentric phase of each repetition, from the first repetition until the MPV was lower than 0.50 $m{\bullet}s^{-1}$. Therefore, the performance in this test was determined as the number of repetitions completed until the first repetition in which the MPV was just less than 0.50 $m{\bullet}s^{-1}$. Following previous studies [38], fatigue test during Pre- and Post-training for each participant was performed with the same absolute load (kg) as a parameter of muscle endurance.

2.4. Resistance training program

All participants conducted a total of 16 training sessions over 8 weeks using only the SQ exercise. All training variables, including relative load (50–80% 1RM), percentage of velocity loss (VL, 15%), number of sets (three), recovery time between sets (4 min) and between sessions (72 h), frequency (2 sessions per week) and even the number of times each load magnitude is used were the same for both experimental groups (Table 2). Thus, the RT program for LP and DUP only differed in how the relative load evolved during the training cycle (Fig. 2). Table 2 shows in detail the descriptive characteristics of the RT program. For both experimental groups, velocity-based RT was used. Thus, relative loads were determined from the load-velocity relationship for the SQ as it has recently been shown that there exists a very close relationship between percentage of 1RM and MPV [32]. Thus, a target MPV to be attained in the first (usually the fastest) repetition of the first exercise set in each training session was used as an estimation of percentage of 1RM, as follows: ~1.16 $m{\bullet}s^{-1}$ (~50% 1RM), ~1.00 $m{\bullet}s^{-1}$ (~60% 1RM), ~0.84 $m{\bullet}s^{-1}$ (~70% 1RM), ~0.75 $m{\bullet}s^{-1}$ (~75% 1RM), and ~0.68 $m{\bullet}s^{-1}$ (~80% 1RM) [43]. Consequently, before starting the first set in each training session, adjustments in the proposed load (kg) were made when needed so that the MPV of the first repetition matched that programmed velocity (± 0.03 $m{\bullet}s^{-1}$). Once the load (kg) was adjusted, it was maintained for the 3 training sets. In addition, volume in each training set was objectively determined through the magnitude of VL

Table. 2
Descriptive characteristics of the squat training program performed by the LP and DUP groups.

Scheduled	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8	Session 9
Sets x Loss (%)	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%
Target MPV ($m{\bullet}s^{-1}$)									
LP	~1.16	~1.16	~1.16	~1.00	~1.00	~1.00	~0.84	~0.84	~0.84
DUP	~1.16	~1.00	~1.16	~0.84	~1.16	~0.84	~1.00	~0.76	~1.00
Actually Performed									
VL (%)									
LP	15.4 ± 0.9	15.7 ± 1.4	16.0 ± 1.1	15.1 ± 1.4	15.9 ± 1.3	16.1 ± 1.3	15.9 ± 1.8	15.6 ± 2.2	16.1 ± 1.3
DUP	15.5 ± 1.1	15.6 ± 1.4	15.5 ± 0.9	15.8 ± 1.4	15.4 ± 1.1	16.0 ± 1.0	15.7 ± 1.2	15.7 ± 2.2	16.1 ± 1.4
N° Rep									
LP	8.0 ± 2.4	8.0 ± 2.6	7.5 ± 2.7	6.2 ± 2.9	5.5 ± 1.7	6.0 ± 2.6	3.9 ± 1.6	4.0 ± 1.5	3.9 ± 1.4
DUP	8.0 ± 2.8	7.2 ± 3.1	8.1 ± 2.2	4.9 ± 2.7	7.8 ± 3.2	4.0 ± 1.5	5.6 ± 2.1	3.2 ± 0.8	5.1 ± 1.6
Reference rep's MPV ($m{\bullet}s^{-1}$)									
LP	1.17 ± 0.03 (~49.5% 1RM)	1.15 ± 0.02 (~50.6% 1RM)	1.15 ± 0.04 (~50.8% 1RM)	1.02 ± 0.03 (~58.8% 1RM)	1.01 ± 0.03 (~59.7% 1RM)	1.02 ± 0.02 (~59.0% 1RM)	0.85 ± 0.02 (~69.8% 1RM)	0.84 ± 0.02 (~69.9% 1RM)	0.85 ± 0.03 (~69.6% 1RM)
DUP	1.17 ± 0.03 (~49.4% 1RM)	1.02 ± 0.02 (~59.1% 1RM)	1.16 ± 0.03 (~50.0% 1RM)	0.85 ± 0.04 (~69.5% 1RM)	1.16 ± 0.03 (~50.1% 1RM)	0.85 ± 0.03 (~69.6% 1RM)	0.76 ± 0.02 (~75.1% 1RM)	1.00 ± 0.03 (~60.0% 1RM)	0.77 ± 0.03 (~74.6% 1RM)
Scheduled	Session 10	Session 11	Session 12	Session 13	Session 14	Session 15	Session 16		
Sets x VL (%)	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%	3 × 15%		
Target MPV ($m{\bullet}s^{-1}$)									
LP	~0.84	~0.76	~0.76	~0.76	~0.68	~0.68	~0.68		
DUP	~0.76	~0.84	~0.68	~0.84	~0.68	~0.76	~0.68		
Actually Performed								Overall	
VL (%)									
LP	15.9 ± 1.3	15.7 ± 2.1	15.9 ± 2.1	15.2 ± 2.1	16.1 ± 2.1	15.9 ± 1.7	15.7 ± 2.3		15.8 ± 1.7
DUP	16.3 ± 2.3	15.8 ± 1.2	15.9 ± 1.8	15.2 ± 1.2	16.0 ± 2.5	15.5 ± 2.5	16.2 ± 1.5		15.8 ± 1.6
N° Rep									
LP	4.2 ± 1.6	3.1 ± 0.9	3.2 ± 1.2	2.8 ± 0.6	2.5 ± 0.7	2.6 ± 0.7	2.4 ± 0.5		4.6 ± 2.6
DUP	3.0 ± 0.9	3.5 ± 1.2	2.5 ± 0.8	3.5 ± 0.9	2.6 ± 0.8	2.8 ± 0.8	2.6 ± 0.9		4.7 ± 2.7
Reference rep's MPV ($m{\bullet}s^{-1}$)									
LP	0.85 ± 0.03 (~69.6% 1RM)	0.77 ± 0.03 (~74.5% 1RM)	0.76 ± 0.03 (~74.8% 1RM)	0.76 ± 0.03 (~75.1% 1RM)	0.70 ± 0.02 (~78.6% 1RM)	0.69 ± 0.04 (~78.9% 1RM)	0.69 ± 0.04 (~79.3% 1RM)		0.89 ± 0.17 (~66.8% 1RM)
DUP	1.00 ± 0.02 (~60.4% 1RM)	0.76 ± 0.02 (~74.8% 1RM)	0.85 ± 0.02 (~69.7% 1RM)	0.68 ± 0.02 (~79.9% 1RM)	0.69 ± 0.03 (~79.4% 1RM)	0.76 ± 0.03 (~74.8% 1RM)	0.68 ± 0.02 (~79.5% 1RM)		0.89 ± 0.17 (~66.9% 1RM)

Data are mean ± SD. Only one exercise (full squat) was used in training. LP: Linear programming group ($n = 16$); DUP: Daily undulatory programming group ($n = 16$); MPV: Mean propulsive velocity attained with the intended load (%1RM); VL: Magnitude of velocity loss expressed as percent loss in mean repetition velocity from the fastest (usually first) to the slowest (last one) repetition of each set; N° Rep: average number of repetitions performed in each training set.

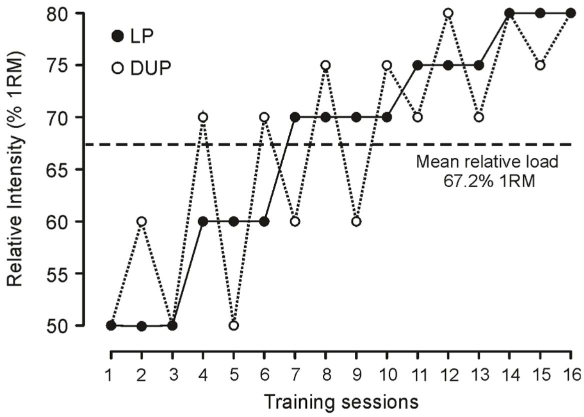

Fig. 2. Evolution of relative load during the training program for linear programming (LP) and daily-undulating programming (DUP).

attained over the set (calculated as the percent loss in MPV from the fastest to the slowest repetition) [30,35]. Thus, the training set was terminated when the prescribed VL limit was reached [30,31]. According to this method, participants of both training groups performed repetitions within the set until reaching 15% VL with respect to the best MPV obtained in such training set. Training sessions was performed on a Smith machine (Multipower Fitness Line, Peroga, Murcia, Spain). All repetitions of all participants during all sessions were recorded using a linear velocity transducer (T-Force system). Furthermore, participants received immediate movement velocity feedback while being encouraged to perform each repetition at maximal intended velocity [42]. During all training sessions, participants carried out a general standardized warm-up which consisted of 5 min of running at a self-selected intensity, 5 min of joint mobilization exercises, followed by 3 sets of progressively faster 30 m running accelerations. The estimate 1RM was calculated for each individual in each training session from the MPV attained against the heaviest load (kg) lifted in the session in order to analyze the evolution of this variable during all training cycle [43,44].

2.5. Statistical analysis

Standard statistical methods were used for the calculation of means, standard deviations (SD). The normality of distribution of the variables at Pre was examined with the Shapiro-Wilk test and the homogeneity of variance across groups (LP vs. DUP) was verified using the Levene's test. A one-way random effects model (model 2,1) ICC was used to determine relative reliability with 95% confidence interval. Absolute reliability was reported using the CV. A 2 (group: LP vs. DUP) x 2 (time: Pre vs. Post) factorial ANOVA with Bonferroni's adjustment was used to analyze the differences between experimental groups. In addition, a 2 (group: LP vs. DUP) x 10 (time: Pre, weeks 1, 2, 3, 4, 5, 6, 7 and 8, and Post) factorial ANOVA with Bonferroni's adjustment was also used to analyze the evolution of CMJ and 1RM during all training cycle. The intra-groups effect sizes (ES_{IN}) were calculated using Hedge's g [45], as follow: g = (mean Post - mean Pre)/Pooled SD. The ES for changes between the LP and DUP (ES_{BE}) in each dependent variable was calculated as follow: g = (mean Pre-Post differences LP) - (mean Pre-Post differences DUP)/Pooled SD. Statistical significance was accepted at $p < 0.05$. The null hypothesis tests were performed using SPSS software version 17.0 (SPSS, Chicago, IL).

3. Results

3.1. Training program

Descriptive characteristics of the training actually performed by LP and DUP are presented in Table 2. There was not any dropout during the study and compliance with the RT program was 100% of all sessions scheduled for both training groups. No significant differences between groups were observed in any of the variables analyzed during training (Table 2). Thus, the best MPV of the first set in each training session (i.e., % 1RM), VL in the set and number of repetitions per set were practically the same for LP and DUP (Table 2). There were no significant differences between groups in the average repetitions performed in different velocity ranges and the average total repetitions performed with the maximal scheduled load in each training session (Table 3). Participants in LP and DUP trained at practically the same average MPV (0.87 ± 0.17 m•s^{-1} vs. 0.87 ± 0.18 m•s^{-1}, respectively).

3.2. Strength and jump performance

All variables analyzed met the criteria of normality and homoscedasticity. There were no significant differences between experimental groups at baseline (Pre) for any of the variables analyzed. Significant "time x group" interaction ($p < 0.05$) for all strength variables (1RM, AV, AV > 1 and AV < 1) and fatigue test were observed in favor of LP group, whereas no significant "time x group" interaction was observed for CMJ performance (Table 4). There were no significant between-groups differences at Post in any variables assessed. Both experimental groups showed significant ($p < 0.001$) Pre-Post change in all variables analyzed, except in fatigue test for DUP (Table 4). The intra-group ESs and percentages of change were greater for LP compared to DUP in all variables measured (Table 4, Figs. 3 and 4).

3.3. Evolution of 1RM and CMJ

The evolutions of the 1RM and CMJ performance throughout the training program are displayed in Fig. 5A and B, respectively. The 1RM value was calculated based on the relationship existing between MPV and%1RM in the SQ exercise. There were no significant 'time x group' interactions and between groups differences in any of the two variables analyzed. The LP group showed significant improvements (p < 0.01 - 0.001) in 1RM from week-3 to Post, whereas for CMJ changes were statistically significant from week-5 to Post. For DUP group, changes in 1RM were significant only at week-6 and Post, while changes in CMJ were significant from week-5 to Post, except at week-7 (Fig. 5). In addition, the intra-group ES were always greater for LP compared to DUP group during all training period.

4. Discussion

The present study described the neuromuscular and functional effects of two RT programs only differing in the relative load distribution (i.e., programming strategy) during the training cycle: LP vs. DUP. To the best of our knowledge, this is the first study in which the valid, reliable and highly sensitive velocity-based RT method was used for comparing two training programming models for: i) monitoring the key

Table. 3
Repetitions performed in different velocity ranges by each training group.

MPV	LP	DUP
> 1.20	0.3 ± 0.6	0.8 ± 2.7
1.20 - 1.10	18.2 ± 7.9	21.3 ± 9.6
1.10 - 1.00	40.4 ± 15.1	40.1 ± 16.9
1.00 - 0.90	42.9 ± 17.1	41.7 ± 18.7
0.90 - 0.80	35.4 ± 15.0	35.5 ± 15.0
0.80 - 0.70	38.7 ± 11.7	36.9 ± 11.1
0.70 - 0.60	30.8 ± 6.5	32.2 ± 11.6
0.60 - 0.50	12.9 ± 5.1	12.9 ± 3.9
0.50 - 0.40	1.5 ± 1.3	1.8 ± 1.4
< 0.40	0.1 ± 0.3	0.1 ± 0.3
REP_{total}	221.1 ± 62.5	223.3 ± 67.6

MPV: Mean propulsive velocity; LP: Linear programming group; DUP: Daily undulatory programming group; REP_{total}: Total number of repetitions.

Table. 4
Changes in selected neuromuscular performance variables from pre- to post-training for each training group.

Variable	LP Pre	Post	ES_{IN} (95% CI)	DUP Pre	Post	ES_{IN} (95% CI)	Changes for LP vs. DUP Δ (%)	ES_{BE} (95% CI)
CMJ	36.7 ± 4.6	41.2 ± 6.0 ***	0.84 (0.45 to 1.23)	36.3 ± 4.9	39.5 ± 4.9 ***	0.65 (0.33 to 0.96)	3.6	0.25 (−0.44 to 0.94)
1RM (kg) †	99.2 ± 22.9	113.3 ± 25.2 ***	0.58 (0.34 to 0.82)	100.7 ± 17.6	109.3 ± 18.3 ***	0.48 (0.20 to 0.75)	5.7	0.26 (−0.44 to 0.96)
AV ($m•s^{-1}$) †	1.08 ± 0.09	1.22 ± 0.10 ***	1.55 (0.86 to 2.24)	1.09 ± 0.09	1.16 ± 0.08 ***	0.94 (0.42 to 1.46)	6.5	0.68 (−0.03 to 1.39)
AV > 1 ($m•s^{-1}$) †	1.37 ± 0.14	1.51 ± 0.16 ***	0.83 (0.42 to 1.24)	1.34 ± 0.08	1.41 ± 0.08 ***	0.87 (0.34 to 1.4)	5.1	0.43 (−0.27 to 1.13)
AV < 1 ($m•s^{-1}$) †	0.73 ± 0.03	0.88 ± 0.08 ***	2.49 (1.45 to 3.57	0.73 ± 0.05	0.84 ± 0.08 ***	1.65 (0.91 to 2.39)	5.5	0.63 (−0.08 to 1.34)
Fatigue test (Rep) †	15.1 ± 6.4	23.4 ± 9.7 ***	1.01 (0.47 to 1.55)	17.5 ± 8.2	20.7 ± 8.0	0.40 (−0.01 to 0.80)	33.7	0.63 (−0.09 to 1.34)

Data are mean ± SD; Pre: initial evaluations; Post: Final evaluations; ES_{IN}: Intra-group effect size; ES_{BE}: Between-groups effect size; CI: Confidence Interval; Δ: Percentage of change; Rep: Number of repetitions completed; LP: Linear programming group ($n = 16$), DUP: Daily undulatory programming group ($n = 16$); 1RM: one-repetition maximum squat strength, AV: average MPV attained against absolute loads common to pre- and post-test in the squat progressive loading test; AV > 1: average MPV attained against absolute loads common to pre- and post-test that were moved faster than 1 $m•s^{-1}$; AV < 1: average MPV attained against absolute loads common to pre- and post-test that were moved slower than 1 $m•s^{-1}$; CMJ: countermovement jump height. Statistically significant "time x group" interaction; † $p < 0.05$. Intra-group significant differences from Pre- to Post-training: *** $p < 0.001$.

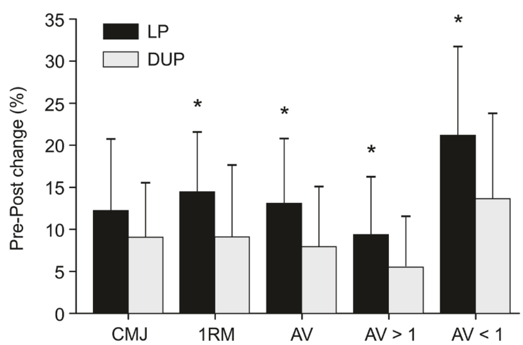

Fig. 3. Relative changes in CMJ and strength variables for linear programming (LP) and daily-undulating programming (DUP). Significant "time x group" interaction: * p 〈 0.05. CMJ: Countermovement jump; 1RM: one-repetition maximum squat strength, AV: average MPV attained against absolute loads common to pre- and post-test in the squat progressive loading test; AV 〉 1: average MPV attained against absolute loads common to pre- and post-test that were moved faster than 1 $m•s^{-1}$; AV < 1: average MPV attained against absolute loads common to pre- and post-test that were moved slower than 1 $m•s^{-1}$. Data are mean ± SD.

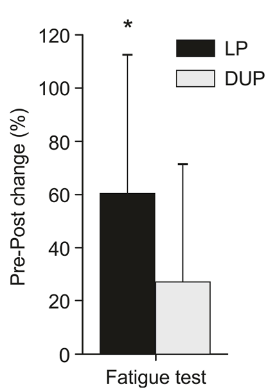

Fig. 4. Relative changes in fatigue test for linear programming (LP) and daily-undulating programming (DUP). Significant "time x group" interaction: * $p < 0.05$. Data are mean ± SD.

training variables (volume and relative load), ii) assessing the weekly performance status (evolution of CMJ and 1RM), and iii) providing valuable parameters related to changes on performance (velocity against light/medium loads and fatigue test). The main findings of the present study were: a) the LP and DUP models significantly improved neuromuscular and functional performance; b) the LP model showed greater ESs and percentages of change in all variables analyzed compared to DUP; and c) the improvements in CMJ and 1RM were earlier and uninterrupted for LP in comparison with DUP.

4.1. Changes in neuromuscular and functional performance

After the 8-week training period, significant "time x group" interactions for all strength variables and fatigue test (Table 4) were found in favor of LP group. In addition, although no significant differences between groups were obtained, the intra- and between-groups ES and percentage of change interpretations suggest greater improvements for LP in comparison with DUP in all variables analyzed (Table 4, Figs. 3 and 4). Comparison of LP vs. DUP strategies has been widely studied in the scientific literature [12-22, 24, 27, 28, 34]. Similarly to the present study, most of these investigations [14,16,18,19,21,22,24,28,34] have shown significant improvements in lower-limb strength and CMJ after both LP and DUP models. However, it is worth noting that the changes found in the current study were achieved with lower relative intensities (50–80% 1RM), volumes (repetitions far from reaching muscle failure: 8–3 repetitions) and number of exercises, than those used in previous studies [14,15,17,24,28,34]. Thus, regardless of the training programming model used, this proposal appears to be a more efficient training method because it induces similar or even greater improvements with a lower degree of fatigue, allowing subsequent technical training in better conditions [46-48].

Although it is already known that LP and DUP induce improvements in physical performance, there is no consensus about what type of programming strategy is more effective in maximizing neuromuscular and functional gains [5,10]. Whereas some investigations have found greater strength improvements by using DUP [12,14-17], others support higher effectiveness of the LP model [21,28]. Indeed, previous systematic reviews and meta-analyses found no significant differences between LP and DUP models, neither for improving upper- and lower-body strength [10] nor muscle hypertrophy [5]. However, our results showed a clear tendency to greater enhancements for LP compared to DUP, as indicated by the "time x groups" interactions (Table 4), the intra-groups ES (0.54 - 2.49 vs. 0.40 - 1.65 for LP and DUP, respectively) and the percentage of change (9.4 - 60.4% vs. 5.5 - 27.2 for LP and DUP, respectively) analyses. These results agree with previous studies [19,27,28], showing greater ES and percentages of change in strength gains in favor of LP compared to DUP model. Authors of the aforementioned studies hypothesized that the contradictory results observed between LP and DUP strategies could be explained by different factors, including study length, volume, relative load, exercises used, or participants' training experience [5,7,10]. Accordingly, it has been suggested that frequent variations of DUP model would provide more effective stimulus than LP to elicit strength improvements and overcome stagnation in participants with prior RT experience [10]. Similarly, it has also been indicated that DUP is more efficient than LP for improving maximal strength in the early phases of a training period [10,12,15]. Nevertheless, in contrast with these assertions, the present study was conducted with trained men (RT experience

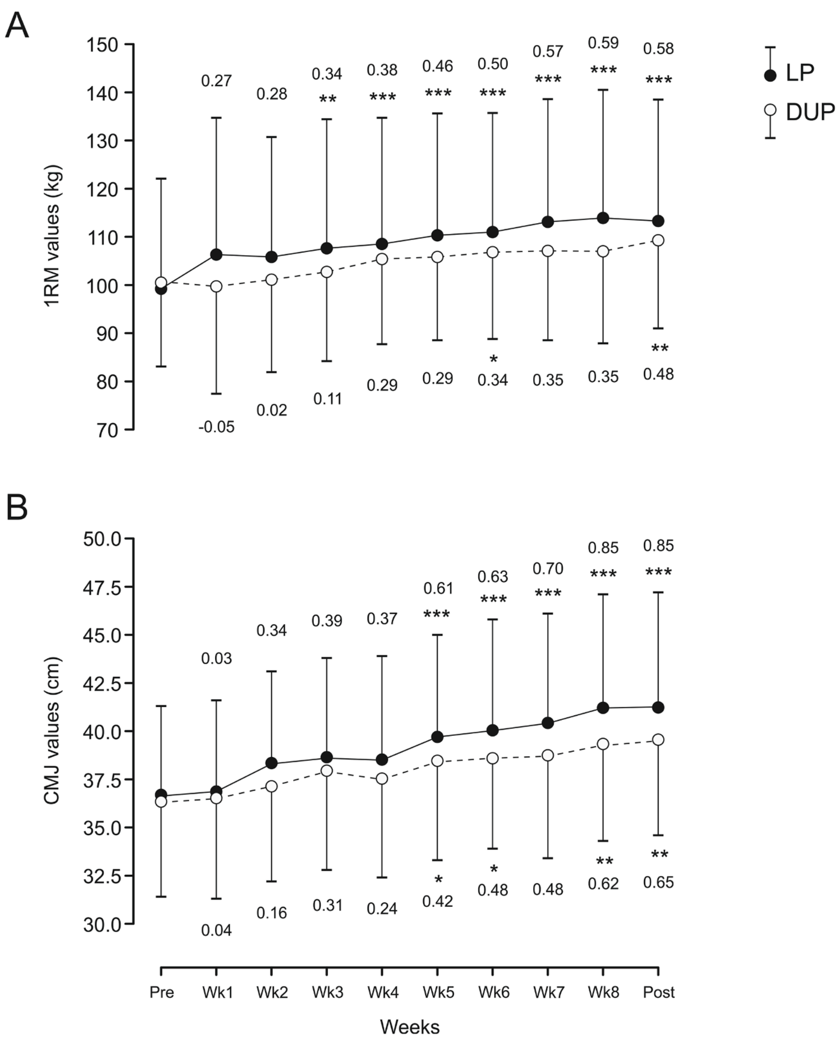

Fig. 5. Evolution of the 1RM strength in the squat exercise and CMJ performance in each training week for linear programming (LP) and daily-undulating programming (DUP). Significant differences with respect to Pre: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Data are mean ± SD. Numbers indicate the intra-group effect size with respect to the Pre-test values.

ranging from 1 to 3 years) and resulted in greater and earlier strength enhancements for LP compared to DUP group. Therefore, it appears that the effects and differences between both programming models could be determined primarily by the relative load and volume used during training program, instead the participants' background in RT.

Traditional LP and DUP comparison studies have conducted RT interventions based on set to muscle failure. However, the level of fatigue generated in sets to failure is different in function of the relative load used. In this regard, achieving muscle failure by using high relative intensities (e.g., 4RM) induces lower degree of fatigue than reaching it by using moderate relative intensities (e.g., 12RM) [31,46-48]. Thus, when high-load sessions (90% 1RM - 4RM) are alternated with lower load sessions (70% 1RM - 12RM), these high-load training sessions could be considered as a "recovery" stimulus, since the degree of fatigue would be lower than that experienced during the lower-load trainings. However, the continuous use of training sessions involving sets with 10–12RM could produce lower increments or even decreases in physical performance [35,44], mainly due to a high accumulation of muscle fatigue in each training session [31,46-48]. In the present study, the degree of fatigue in each training session was low (15% VL) [46], favoring a significant and continuous improvement throughout the training cycle.

In addition to 1RM (kg) and vertical jump tests commonly used for evaluating the efficacy of LP and DUP strategies [10], the present study provided information about changes induced against light (AV>1) and moderate loads (AV<1). This analysis showed that, although both experimental groups performed the same number of repetitions within the different velocity ranges (Table 3), LP model produced greater improvements at light (Δ: 9.4%; ES_{IN}: 0.83) and moderate loads (Δ: 21.2%; ES_{IN}: 2.49) than DUP (Table 4 and Fig. 3). Also, regarding the fatigue test, LP group showed higher and significant gains compared to DUP (Table 4 and Fig. 4). These results are in contrast with previous studies comparing LP vs. DUP models [18, 24], which found similar increases in local muscular endurance between both programming approaches. It is likely that the largest increments in the fatigue test obtained for LP group are due, at least partially, to the greatest increase in maximal strength (1RM and AV<1) shown by this group compared to DUP. On this matter, an increase in the 1RM would allow a greater number of repetitions against the same absolute load, as that absolute load would represent a lower relative intensity and it will be moved faster in the first repetition [38].

Concerning the evolutions of the 1RM and CMJ during the training program, our results showed that changes in both variables were superior, earlier and uninterrupted in LP group compared to DUP (Fig. 5). These findings seem to call into question the two main disadvantages traditionally related to LP model: i) the fatigue state produced by the continuous (although not daily) intensity increment; and ii) the stagnation generated by lengthening the training in a specific loading zone [11]. Indeed, previous studies [12, 15] found that DUP model resulted in muscle strength improvements in the initial weeks of the RT period with no strength gains for LP model until later stages of the intervention [10]. As indicated above, discrepancies with our results could be due to the differences in relative loads and especially to the training volume (i.e., distance to muscle failure in each training set) used or the degree of fatigue reached. In the present study, not conducting sets to muscle failure would have allowed greater adaptations from the beginning of the intervention. Specifically, in order to maximize the effectiveness of LP and DUP models and reduce fatigue in each training session, the current study used a 15% VL. This VL has been proved as an effective and efficient training stimulus for inducing gains in strength, muscle endurance and CMJ performance [38,44]. These continuous 1RM and CMJ evaluations have been previously used weekly, even daily, for monitoring recovery status [47,48] and for analyzing the effectiveness of training interventions [44]. However, to the best of our knowledge, no previous study has measured weekly the changes on maximal strength (1RM) and functional performance (CMJ) for comparing LP and DUP strategies. This simple and quick performance analysis could be used as an interesting method for: i) knowing the effectiveness of each programming model in different stages of the training intervention (i.e., short-term to long-term); and ii) adjusting the load programmed in function of 1RM improvements or decrements.

4.2. Velocity-based RT programs analysis

One of the main strengths of the current study was the use of movement velocity as an objective tool for monitoring, adjusting and quantifying the key variables during RT. The velocity-based RT method allowed us to improve some methodological aspects of traditional studies comparing LP and DUP models. Firstly, the volume and relative load of both groups were equated (Table 2 and 3). Thus, we solved one of the major methodological problems of studies in which different programming strategies have been compared [12,16,18,20,21,27,29]. Volume was equated by using the VL strategy [30,33], whereas the load-velocity relationship of the SQ exercise was used for monitoring the relative load [32]. Studies comparing LP and DUP models have commonly established the training volume using a fixed number of sets and repetitions per set against a determined relative load (e.g., 3 × 12 reps) [16,20-22,28]. Nevertheless, this pre-fixed number of repetitions per set could represent a different level of effort for 2 subjects, due to the fact that the number of repetitions that can be completed against a given relative load presents a large variability between individuals [30,33]. On the other hand, percentage of 1RM or *n*RM (e.g., 75% 1RM or 12RM) methods have been generally used for programming the relative load [14,15,17,24,28,34]. However, as shown in Fig. 5 of the present study, 1RM performance measured at the beginning of an intervention will be considerably modified over the training weeks due to the strength enhances and/or fatigue processes that occur in the athletes' performance [35]. To the best of our knowledge, although previous studies have used the velocity-based RT approach to analyze the effect of different variables such as the intraset training volume [35, 38], range of motion [49] or contraction velocity [42], to date no investigation has used this training approach for comparing different programming strategies. Thus, this is the first study comparing LP and DUP configurations, which objectively verified that both experimental groups carried out the same training program. As observed in Table 2, the only difference between the RT programs was the way in which the relative loads were distributed along the training program. However, further studies are needed in order to extend the knowledge about the different programming models by using VL and load-velocity relationship for monitoring volume and relative load, respectively, during RT.

4.3. Limitations

The current study presents some limitations that need to be addressed. One of the main limitations of the present study was that, although participants were required not to participate in any other type of strenuous physical activity during the time this investigation lasted, the possible practice of exercise outside the training protocol was not really controlled. However, since the groups were randomly assigned, the possible influence of these physical activities on changes in neromuscular performance was probably the same in both experimental groups. Therefore, it could be assumed that the strength and jump gains obtained could have been different if the subjects had not performed any other additional physical activity, but it is also likely that this factor did not affect the comparison between LP and DUP approaches. On the other hand, in the present study only two RT programming models have been explored and compared. Thus, further studies are needed to clarify the programming approach inducing the most beneficial effects on neuromuscular performance. Another limitation was that the study did not include a control group and as a consequence the influence of environmental variables cannot be ruled out.

5. Conclusions

The current study demonstrated that both LP and DUP models are effective training programming for improving strength, jump and muscle endurance performance, by following a prolonged velocity-based RT intervention. Nevertheless, the LP model appears to produce greater, earlier, and uninterrupted enhances in all variables analyzed in comparison with DUP strategy.

6. Practical applications

Results of the present study showed that both programming models were effectives for improving neuromuscular and functional performance. Thus, LP and DUP programming strategies could be chosen by practitioners based on individual possibilities/preferences. However, increases in strength and jump performance were greater, earlier and uninterrupted in the LP group compared to DUP. Therefore, when a RT program is based on moderate loads, a low number of repetitions or low percentage VL in the set and lifting loads at maximal intended velocity in each repetition, the LP model should preferably be used for maximizing improvements of physical performance.

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Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

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References

[1] W.J. Kraemer, N.A. Ratamess, Fundamentals of resistance training: progression and exercise prescription, Med Sci Sports Exerc 36 (2004) 674–688, 00005768-200404000-00017.

[2] B.A. Spiering, W.J. Kraemer, J.M. Anderson, L.E. Armstrong, B.C. Nindl, J.S. Volek, et al., Resistance exercise biology: manipulation of resistance exercise programme variables determines the responses of cellular and molecular signalling pathways, Sports Med 38 (2008) 527–540, 3871.

[3] J.J. Gonzalez-Badillo, D. Rodriguez-Rosell, L. Sanchez-Medina, E.M. Gorostiaga, F. Pareja-Blanco, Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training, Eur J Sport Sci (2014) 1–10, https://doi.org/10.1080/17461391.2014.905987.

[4] B.J. Schoenfeld, D. Ogborn, J.W. Krieger, Effects of resistance training frequency on measures of muscle hypertrophy: a systematic review and meta-analysis, Sports Med 46 (2016) 1689–1697, https://doi.org/10.1007/s40279-016-0543-8.

[5] J. Grgic, P. Mikulic, H. Podnar, Z. Pedisic, Effects of linear and daily undulating periodized resistance training programs on measures of muscle hypertrophy: a systematic review and meta-analysis, PeerJ 5 (2017) e3695, https://doi.org/10.7717/peerj.3695.
[6] H. Hartmann, K. Wirth, M. Keiner, C. Mickel, A. Sander, E. Szilvas, Short-term periodization models: effects on strength and speed-strength performance, Sports Med 45 (2015) 1373–1386, https://doi.org/10.1007/s40279-015-0355-2.
[7] T.D. Williams, D.V. Tolusso, M.V. Fedewa, M.R. Esco, Comparison of periodized and non-periodized resistance training on maximal strength: a meta-analysis, Sports Med 47 (2017) 2083–2100, https://doi.org/10.1007/s40279-017-0734-y.
[8] A.J. Cunanan, B.H. DeWeese, J.P. Wagle, K.M. Carroll, R. Sausaman, W. G. Hornsby, et al., The general adaptation syndrome: a foundation for the concept of periodization, Sports Med 48 (2018) 787–797, https://doi.org/10.1007/s40279-017-0855-3.
[9] M.R. Rhea, B.L. Alderman, A meta-analysis of periodized versus nonperiodized strength and power training programs, Res Q Exerc Sport 75 (2004) 413–422, https://doi.org/10.1080/02701367.2004.10609174.
[10] S.K. Harries, D.R. Lubans, R. Callister, Systematic review and meta-analysis of linear and undulating periodized resistance training programs on muscular strength, J Strength Cond Res 29 (2015) 1113–1125, https://doi.org/10.1519/JSC.0000000000000712.
[11] J.W. Evans, Periodized resistance training for enhancing skeletal muscle hypertrophy and strength: a mini-review, Front Physiol 10 (2019) 13, https://doi.org/10.3389/fphys.2019.00013.
[12] M.R. Rhea, S.D. Ball, W.T. Phillips, L.N. Burkett, A comparison of linear and daily undulating periodized programs with equated volume and intensity for strength, J Strength Cond Res 16 (2002) 250–255.
[13] L.C. Caldas, L. Guimarães-Ferreira, M.J. Duncan, A.S. Leopoldo, A.P.L. Leopoldo, W Lunz, Traditional vs. undulating periodization in the context of muscular strength and hypertrophy: a meta-analysis, Int J Sport Sci 6 (2016) 219–229, https://doi.org/10.5923/j.sports.20160606.04.
[14] M.D. Peterson, D.J. Dodd, B.A. Alvar, M.R. Rhea, M. Favre, Undulation training for development of hierarchical fitness and improved firefighter job performance, J Strength Cond Res 22 (2008) 1683–1695, https://doi.org/10.1519/JSC.0b013e31818215f4.
[15] A.G. Monteiro, M.S. Aoki, A.L. Evangelista, D.A. Alveno, G.A. Monteiro, C. Picarro Ida, et al., Nonlinear periodization maximizes strength gains in split resistance training routines, J Strength Cond Res 23 (2009) 1321–1326, https://doi.org/10.1519/JSC.0b013e3181a00f96.
[16] J. Prestes, A.B. Frollini, C. de Lima, F.F. Donatto, D. Foschini, R. de Cassia Marqueti, et al., Comparison between linear and daily undulating periodized resistance training to increase strength, J Strength Cond Res 23 (2009) 2437–2442, https://doi.org/10.1519/JSC.0b013e3181c03548.
[17] R. Simao, J. Spineti, B.F. de Salles, T. Matta, L. Fernandes, S.J. Fleck, et al., Comparison between nonlinear and linear periodized resistance training: hypertrophic and strength effects, J Strength Cond Res 26 (2012) 1389–1395, https://doi.org/10.1519/JSC.0b013e318231a659.
[18] M.R. Rhea, W.T. Phillips, L.N. Burkett, W.J. Stone, S.D. Ball, B.A. Alvar, et al., A comparison of linear and daily undulating periodized programs with equated volume and intensity for local muscular endurance, J Strength Cond Res 17 (2003) 82–87, https://doi.org/10.1519/1533-4287(2003)017.
[19] T.W. Buford, S.J. Rossi, D.B. Smith, A.J. Warren, A comparison of periodization models during nine weeks with equated volume and intensity for strength, J Strength Cond Res 21 (2007) 1245–1250, https://doi.org/10.1519/R-20446.1.
[20] H. Hartmann, A. Bob, K. Wirth, D. Schmidtbleicher, Effects of different periodization models on rate of force development and power ability of the upper extremity, J Strength Cond Res 23 (2009) 1921–1932, https://doi.org/10.1519/JSC.0b013e3181b73c69.
[21] J.R. Hoffman, N.A. Ratamess, M. Klatt, A.D. Faigenbaum, R.E. Ross, N. M. Tranchina, et al., Comparison between different off-season resistance training programs in Division III American college football players, J Strength Cond Res 23 (2009) 11–19, https://doi.org/10.1519/jsc.0b013e3181876a78.
[22] L.Y. Kok, P.W. Hamer, D.J. Bishop, Enhancing muscular qualities in untrained women: linear versus undulating periodization, Med Sci Sports Exerc 41 (2009) 1797–1807, https://doi.org/10.1249/MSS.0b013e3181a154f3.
[23] S.K. Harries, D.R. Lubans, R. Callister, Comparison of resistance training progression models on maximal strength in sub-elite adolescent rugby union players, J Sci Med Sport 19 (2016) 163–169, https://doi.org/10.1016/j.jsams.2015.01.007.
[24] E. Franchini, B.M. Branco, M.F. Agostinho, M. Calmet, R. Candau, Influence of linear and undulating strength periodization on physical fitness, physiological, and performance responses to simulated judo matches, J Strength Cond Res 29 (2015) 358–367, https://doi.org/10.1519/JSC.0000000000000460.
[25] B. Ullrich, T. Pelzer, M. Pfeiffer, Neuromuscular effects to 6 weeks of loaded countermovement jumping with traditional and daily undulating periodization, J Strength Cond Res 32 (2018) 660–674, https://doi.org/10.1519/JSC.0000000000002290.
[26] T. Pelzer, B. Ullrich, M. Pfeiffer, Periodization effects during short-term resistance training with equated exercise variables in females, Eur J Appl Physiol 117 (2017) 441–454, https://doi.org/10.1007/s00421-017-3544-x.
[27] J.R. Hoffman, M. Wendell, J. Cooper, J. Kang, Comparison between linear and nonlinear in-season training programs in freshman football players, J Strength Cond Res 17 (2003) 561–565, https://doi.org/10.1519/1533-4287(2003)017.
[28] C. de Lima, D.A. Boullosa, A.B. Frollini, F.F. Donatto, R.D. Leite, P.R. Gonelli, et al., Linear and daily undulating resistance training periodizations have differential beneficial effects in young sedentary women, Int J Sports Med 33 (2012) 723–727, https://doi.org/10.1055/s-0032-1306324.
[29] S. Ahmadizad, S. Ghorbani, M. Ghasemikaram, M. Bahmanzadeh, Effects of short-term nonperiodized, linear periodized and daily undulating periodized resistance training on plasma adiponectin, leptin and insulin resistance, Clin Biochem 47 (2014) 417–422, https://doi.org/10.1016/j.clinbiochem.2013.12.019.
[30] J.J. Gonzalez-Badillo, J.M. Yanez-Garcia, R. Mora-Custodio, D. Rodriguez-Rosell, Velocity loss as a variable for monitoring resistance exercise, Int J Sports Med 38 (2017) 217–225, https://doi.org/10.1055/s-0042-120324.
[31] L. Sanchez-Medina, J.J. Gonzalez-Badillo, Velocity loss as an indicator of neuromuscular fatigue during resistance training, Med Sci Sports Exerc 43 (2011) 1725–1734, https://doi.org/10.1249/MSS.0b013e318213f880.
[32] L. Sánchez-Medina, J.G. Pallarés, C.E. Pérez, R. Morán-Navarro, J.J. González-Badillo, Estimation of relative load from bar velocity in the full back squat exercise, Sports Med Int Open 1 (2017) E80–E88, https://doi.org/10.1055/s-0043-102933.
[33] D. Rodriguez-Rosell, J.M. Yanez-Garcia, L. Sanchez-Medina, R. Mora-Custodio, J. J. Gonzalez-Badillo, Relationship between velocity loss and repetitions in reserve in the bench press and back squat exercises, J Strength Cond Res (2019), https://doi.org/10.1519/JSC.0000000000002881.
[34] F. Miranda, R. Simao, M. Rhea, D. Bunker, J. Prestes, R.D. Leite, et al., Effects of linear vs. daily undulatory periodized resistance training on maximal and submaximal strength gains, J Strength Cond Res 25 (2011) 1824–1830, https://doi.org/10.1519/JSC.0b013e3181e7ff75.
[35] F. Pareja-Blanco, D. Rodriguez-Rosell, L. Sanchez-Medina, J. Sanchis-Moysi, C. Dorado, R. Mora-Custodio, et al., Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations, Scand J Med Sci Sports 27 (2017) 724–735, https://doi.org/10.1111/sms.12678.
[36] M. Izquierdo, J. Ibanez, J.J. Gonzalez-Badillo, K. Hakkinen, N.A. Ratamess, W. J. Kraemer, et al., Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains, J Appl Physiol 100 (2006) 1647–1656, https://doi.org/10.1152/japplphysiol.01400.2005.
[37] M. Izquierdo-Gabarren, R Gonzalez De Txabarri Exposito, J. Garcia-pallares, L. Sanchez-medina, E.S. De Villarreal, M Izquierdo, Concurrent endurance and strength training not to failure optimizes performance gains, Med Sci Sports Exerc 42 (2010) 1191–1199, https://doi.org/10.1249/MSS.0b013e3181c67eec.
[38] D. Rodriguez-Rosell, J.M. Yanez-Garcia, R. Mora-Custodio, F. Pareja-Blanco, A. G. Ravelo-Garcia, J. Ribas-Serna, et al., Velocity-based resistance training: impact of velocity loss in the set on neuromuscular performance and hormonal response, Appl Physiol Nutr Metab (2020), https://doi.org/10.1139/apnm-2019-0829.
[39] T. Pelzer, B. Ullrich, S. Endler, C. Rasche, M. Pfeiffer, A biomechanical comparison of countermovement performance after short-term traditional and daily-undulated loaded vertical jump training, J Sports Sci 36 (2018) 1816–1826, https://doi.org/10.1080/02640414.2017.1422627.
[40] D. Rodriguez-Rosell, R. Mora-Custodio, F. Franco-Marquez, J.M. Yanez-Garcia, J. J. Gonzalez-Badillo, Traditional vs. sport-specific vertical jump tests: reliability, validity, and relationship with the legs strength and sprint performance in adult and teen soccer and basketball players, J Strength Cond Res 31 (2017) 196–206, https://doi.org/10.1519/JSC.0000000000001476.
[41] J.G. Claudino, J. Cronin, B. Mezencio, D.T. McMaster, M. McGuigan, V. Tricoli, et al., The countermovement jump to monitor neuromuscular status: a meta-analysis, J Sci Med Sport 20 (2017) 397–402, https://doi.org/10.1016/j.jsams.2016.08.011.
[42] F. Pareja-Blanco, D. Rodriguez-Rosell, L. Sanchez-Medina, E.M. Gorostiaga, J. J. Gonzalez-Badillo, Effect of movement velocity during resistance training on neuromuscular performance, Int J Sports Med 35 (2014) 916–924, https://doi.org/10.1055/s-0033-1363985.
[43] L. Sanchez-Medina, J.G. Pallares, C.E. Perez, R. Moran-Navarro, J.J. Gonzalez-Badillo, Estimation of relative load from bar velocity in the full back squat exercise, Sports Med Int Open 1 (2017) E80–E88, https://doi.org/10.1055/s-0043-102933.
[44] F. Pareja-Blanco, L. Sanchez-Medina, L. Suarez-Arrones, J.J. Gonzalez-Badillo, Effects of velocity loss during resistance training on performance in professional soccer players, Int J Sports Physiol Perform 12 (2017) 512–519, https://doi.org/10.1123/ijspp.2016-0170.
[45] L.V. Hedges, I.O. Olkin, Statistical Methods For Meta-Analysis, Academic Press, Orlando, FL, 1985.
[46] D. Rodriguez-Rosell, J.M. Yanez-Garcia, J. Torres-Torrelo, R. Mora-Custodio, M. C. Marques, J.J. Gonzalez-Badillo, Effort index as a novel variable for monitoring the level of effort during resistance exercises, J Strength Cond Res 32 (2018) 2139–2153, https://doi.org/10.1519/JSC.0000000000002629.
[47] R. Moran-Navarro, C.E. Perez, R. Mora-Rodriguez, E. de la Cruz-Sanchez, J. J. Gonzalez-Badillo, L. Sanchez-Medina, et al., Time course of recovery following resistance training leading or not to failure, Eur J Appl Physiol 117 (2017) 2387–2399, https://doi.org/10.1007/s00421-017-3725-7.
[48] F. Pareja-Blanco, D. Rodriguez-Rosell, P. Aagaard, L. Sanchez-Medina, J. Ribas-Serna, R. Mora-Custodio, et al., Time course of recovery from resistance exercise with different set configurations, J Strength Cond Res (2018), https://doi.org/10.1519/JSC.0000000000002756.
[49] J.G. Pallares, A.M. Cava, J. Courel-Ibanez, J.J. Gonzalez-Badillo, R. Moran-Navarro, Full squat produces greater neuromuscular and functional adaptations and lower pain than partial squats after prolonged resistance training, Eur J Sport Sci (2019) 1–10, https://doi.org/10.1080/17461391.2019.1612952.